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Data Visualization Final Report: Trends in US Manufacturing

I. Problem Statement

Study examining regional trends in USA manufacturing employment (growth in Florida vs. declines in New York), including the impacts of economic downturns in 2008 and 2019, and potential correlations with automation adoption.

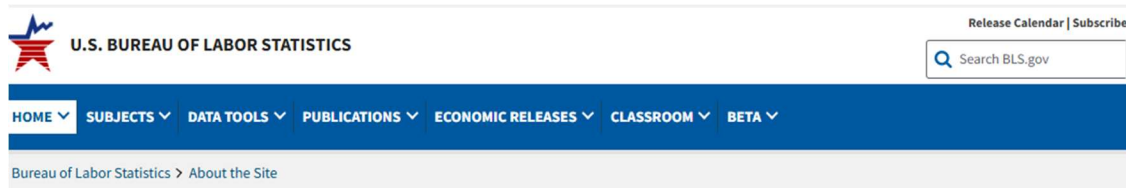
II. Motivation

I am interested in exploring this data because manufacturing is a key part of the US economy that employs millions and influences quality of life. Manufacturing in the USA is personally interesting to me as I begin my career, having previously worked in a factory, and with the many applications of data science in industry from predictive maintenance and computer vision for robotics to business analytics.

I am curious to know more about how the industry is trending and what factors may affect it. Particularly, I'd like to look for correlations with other datasets. Robotics has the potential to radically change the manufacturing industry, and it would be interesting to understand the impact on the job market. These understandings could guide policymakers on decisions such as Automation and AI.

III. Data Sources

1. BLS.gov Employment Data in Manufacturing



Linking and Copyright Information

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Figure 1 BLS.gov Copyright Information

2. Census.gov Population Data

Notes: The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product (Data Management System (DMS) number: P-6000042 and P-7501659. Disclosure Review Board (DRB) approval number: CBDRB-FY25-0466).

Figure 2 Census.gov Population Data

3. Census.gov Robotics Datasets

<https://www.census.gov/library/publications/2025/econ/2022-ec-robotic-equipment.html>

<https://www.census.gov/library/publications/2019/econ/2018-robotic-equipment.html#tables>

Getting the Time Series Data from BLS.gov

I scraped manufacturing employment time series data from <https://api.bls.gov/publicAPI/v2/timeseries/data/> for my initial survey.

```
bls.py x
src > Data Gathering > bls.gov > employment > bls.py > ...
1
2 import requests
3 import pandas as pd
4 import time
5 import os
6 from pathlib import Path
7
8
9 class StateEmployment:
10     def __init__(self, filename='output.csv'):
11         self.base_url = "https://api.bls.gov/publicAPI/v2/timeseries/data/"
12         self.apikey = os.environ.get('BLS_API_KEY')
13         self.states = pd.read_csv('sm.state.txt', delimiter='\t')
14         self.area_codes = pd.read_csv('sm.area.txt', delimiter='\t')
15         self.industries = pd.read_csv('sm.industry.txt', delimiter='\t')
16         self.fn = filename
17     > self.us_states = { ...
75
76     > def fetch_data(self, series_ids: list, endyear=2026): ...
99
100    > def build_series_id(self, state_code: str, area_code: str, ...
108
109    > def get_statecode_from_area(self, area_name: str) -> str: ...
118
119    > def get_industry_code(self, keyword: str) -> list: ...
123
124    > def make_series(self, keyword: str = "Manufacturing", data_type:str = '01') -> list: ...
149
150    > def subset_series(self, series:list, chunk_len:int=50) -> list: ...
163
164    > def download(self, series:list): ...
180
181
182    > def save_data(self, json): ...
213
214    > def extract_area_from_series_id(self, series_id): ...
222
223
224
225 if __name__ == '__main__':
226     script_dir = Path(__file__).parent.resolve()
227     DATA_OUT_DIR = (script_dir / '../data/bls.gov/employment/raw').resolve()
228     os.makedirs(DATA_OUT_DIR, exist_ok=True)
229     data_types = ['01', '03', '06', '07', '08', '30']
230
231     for dt in data_types:
232         fn = os.path.join(DATA_OUT_DIR, dt+'.csv')
233         s = StateEmployment(filename=fn)
234         series = s.make_series(data_type=dt)
235         print("Total series:",len(series))
236         s.download(series)
237         print("Done scraping datatype:", dt)
238
```

Figure 3 Scraping Timeseries Data from BLS.gov

Available timeseries data started in 2007, ended in 2025, and there was data for 392 unique areas. I cleaned the data and created a state feature that extracts the state code from the area which left me with the features below:

```
df.head(3)
```

[11]

	series_id	state	area	year	period	month	total_workers
0	SMU152798030000000001	Hawaii	Kahului-Wailuku, HI	2025	M12	December	1100.0
1	SMU152798030000000001	Hawaii	Kahului-Wailuku, HI	2025	M11	November	1100.0
2	SMU152798030000000001	Hawaii	Kahului-Wailuku, HI	2025	M10	October	1100.0

Processing to get per capita data

I downloaded the population data as a CSV, matched the states to the employment data, and used it to adjust the years 2020-2025 per capita.

```
rows = []

for state in state_set:
    subset = df[df.state == state]
    year_2020 = subset[subset.year == 2020]
    population = df_pop[df_pop.STATE_NAME == state].iloc[0].POPESTIMATE2020
    year_2020['per_capita'] = year_2020.value.astype('int') / population
    rows.append(year_2020)

    year_2021 = subset[subset.year == 2021]
    population = df_pop[df_pop.STATE_NAME == state].iloc[0].POPESTIMATE2021
    year_2021['per_capita'] = year_2021.value.astype('int') / population
    rows.append(year_2021)

    year_2022 = subset[subset.year == 2022]
    population = df_pop[df_pop.STATE_NAME == state].iloc[0].POPESTIMATE2022
    year_2022['per_capita'] = year_2022.value.astype('int') / population
    rows.append(year_2022)

    year_2023 = subset[subset.year == 2023]
    population = df_pop[df_pop.STATE_NAME == state].iloc[0].POPESTIMATE2023
    year_2023['per_capita'] = year_2023.value.astype('int') / population
    rows.append(year_2023)

    year_2024 = subset[subset.year == 2024]
    population = df_pop[df_pop.STATE_NAME == state].iloc[0].POPESTIMATE2024
    year_2024['per_capita'] = year_2024.value.astype('int') / population
    rows.append(year_2024)

    year_2025 = subset[subset.year == 2025]
    population = df_pop[df_pop.STATE_NAME == state].iloc[0].POPESTIMATE2025
    year_2025['per_capita'] = year_2025.value.astype('int') / population
    rows.append(year_2025)
```

Processing to get percent change

I also processed the data to calculate the percentage change over five-year increments for: 2020-2024, 2015-2019, and 2010-2014.

```
analyze_perc_change.py X
src > Processing > bls.gov > analyze_perc_change.py > ...
1 import pandas as pd
2 from pathlib import Path
3 import os
4
5
6 script_dir = Path(__file__).parent.resolve()
7 fn = (script_dir / '../data/bls.gov/employment/raw/01.csv').resolve()
8 df = pd.read_csv(fn, delimiter='\t')
9 fn_out = (script_dir / '../data/bls.gov/employment/processed/01_5yr_perc_change.csv')
10 dir_processed, tail = os.path.split(fn_out)
11 os.makedirs(dir_processed, exist_ok=True)
12
13
14 percent_change_array = []
15 area_set = set(df.area)
16 for _area in area_set:
17     sub_set = df[df.area == _area]
18     min_year = sub_set.year.min()
19     max_year = sub_set.year.max()
20
21     year_spans = [span for y in range(max_year, min_year - 1, -5)
22                  | if len(span := list(range(y, max(y - 5, min_year - 1), -1))) == 5]
23
24     for span in year_spans:
25         state = sub_set.iloc[1].state
26         area = _area
27         new_year = span[0]
28         old_year = span[-1]
29
30         new_number = sub_set[sub_set.year == new_year].iloc[0].value
31         old_number = sub_set[sub_set.year == old_year].iloc[0].value
32
33         perc_change = (new_number - old_number) / old_number
34         #print('-'*13)
35         #print(state, area)
36         #print(new_year, old_year)
37         #print(new_number, old_number, perc_change)
38
39         percent_change_array.append([state, area, old_year, new_year, old_number, new_number, perc_change])
40
41
42 new_df = pd.DataFrame(percent_change_array)
43 new_df.columns = ['state', 'area', 'old_year', 'new_year', 'old_number', 'new_number', 'perc_change']
44 print(new_df.head())
45 print(new_df.shape)
46 new_df.to_csv(fn_out)
47 #print(_state, sub_set.shape)
```

Figure 4 Analysis Code

```
df2 = pd.read_csv("../data/bls.gov/employment/processed/01_5yr_perc_change.csv", index_col=0)
df2.head()
```

✓ 0.0s

	state	area	old_year	new_year	old_number	new_number	perc_change
0	South Carolina	Spartanburg, SC	2021	2025	37.2	41.0	0.102151
1	South Carolina	Spartanburg, SC	2016	2020	29.9	37.0	0.237458
2	South Carolina	Spartanburg, SC	2011	2015	26.4	29.0	0.098485
3	Texas	Dallas-Fort Worth-Arlington, TX	2021	2025	292.6	307.6	0.051265
4	Texas	Dallas-Fort Worth-Arlington, TX	2016	2020	266.6	281.5	0.055889

Figure 5 New Dataset

Gathering Robotics Data

Robotics data was scraped manually. Census.gov started collecting Capital Expenditure data in 2018 and has data for each year until after 2022. The 2022 dataset has more granular details including breakdowns of CE by State.

IV. Visuals

Overview of Employment:

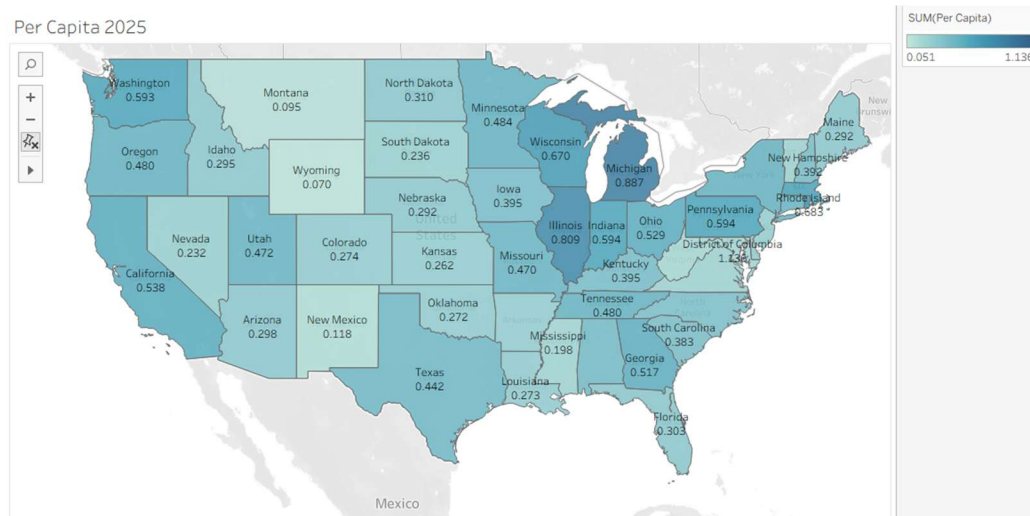


Figure 6 2025 Per Capita Choropleth

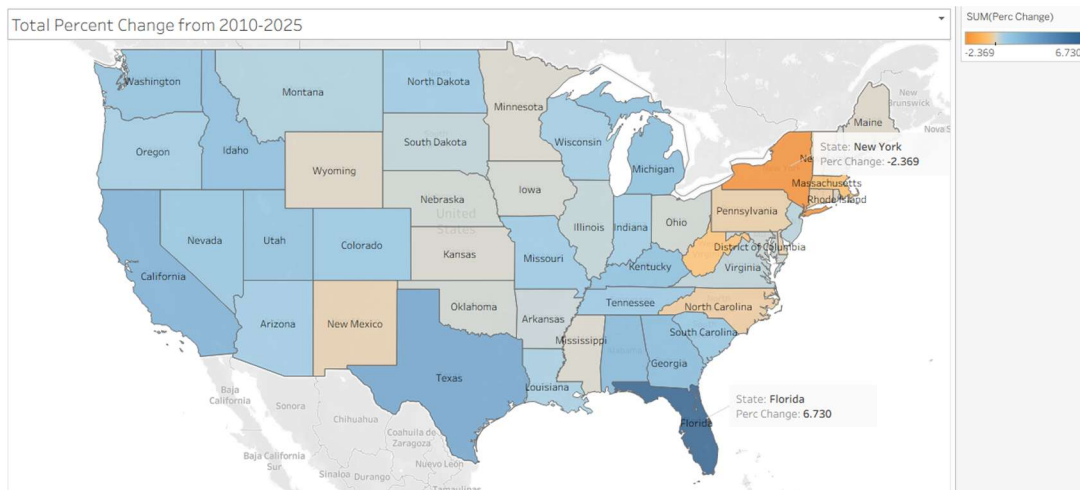


Figure 7 This shows trends by state

This shows an overview of which states employ the most and which states had the greatest change in employment from 2010 – 2025. New York, Massachusetts, West Virginia, and North Carolina all show significant decline. Florida shows the biggest positive change.

Analysis of Eastern Trends:

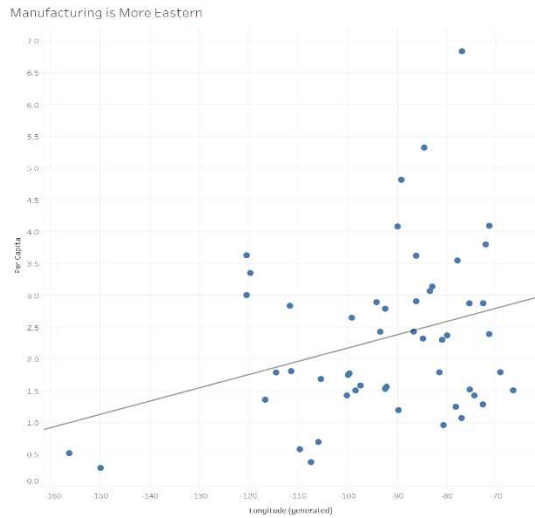


Figure 8 Scatter Plot of Employment Vs Longitude

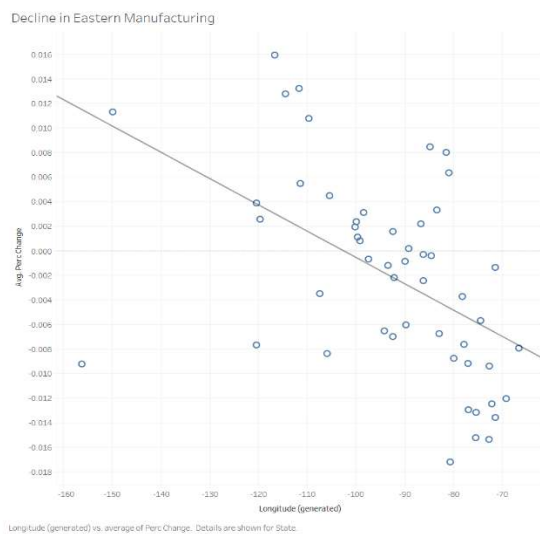


Figure 9 Scatter Plot % Change Vs Longitude

This shows there is a correlation between decline and how eastern a state is. This includes states that may be outliers like Florida and New York. It also shows a correlation that eastern states tend to employ more in manufacturing.

Investment Analysis

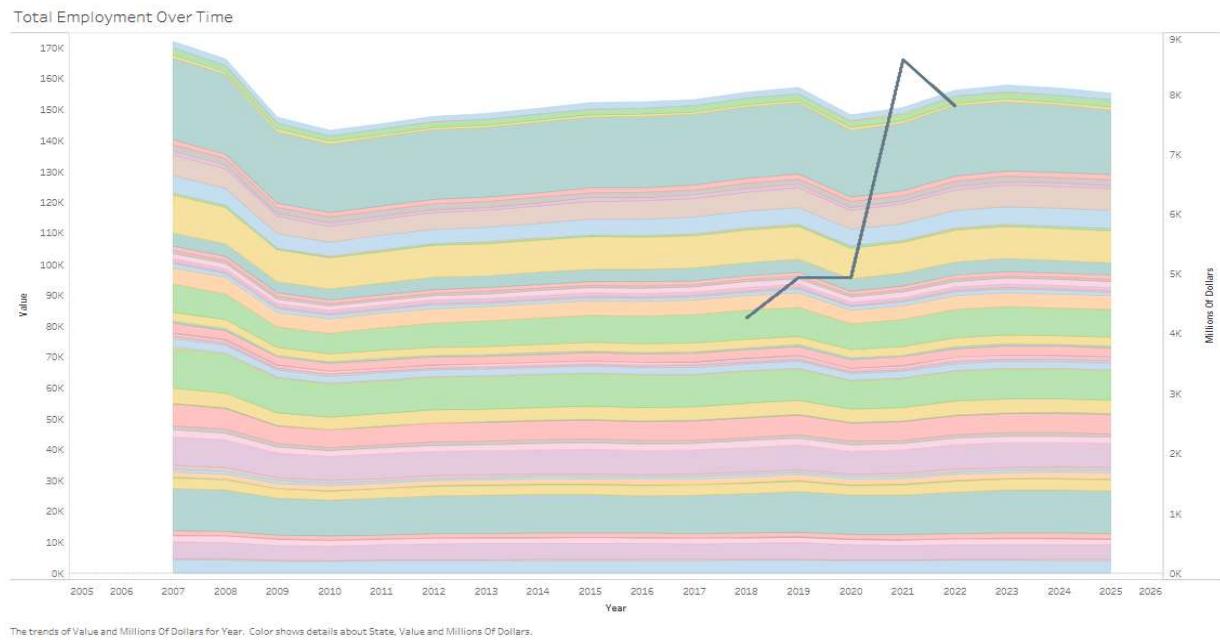


Figure 10 Dual Axis Time Series Analysis of Total Employment & Robotics Capital Expenditure

Time series analysis shows that robotics investment spiked during the decline in employment from Covid. This shows that businesses likely attempted to offset declines with investments in robotics. It is not apparent if it worked, but it did not hurt the recovery process after Covid. However, the larger overall trend shows a lack of recovery in total employment since the 2008 financial crisis.

Investment by Employment

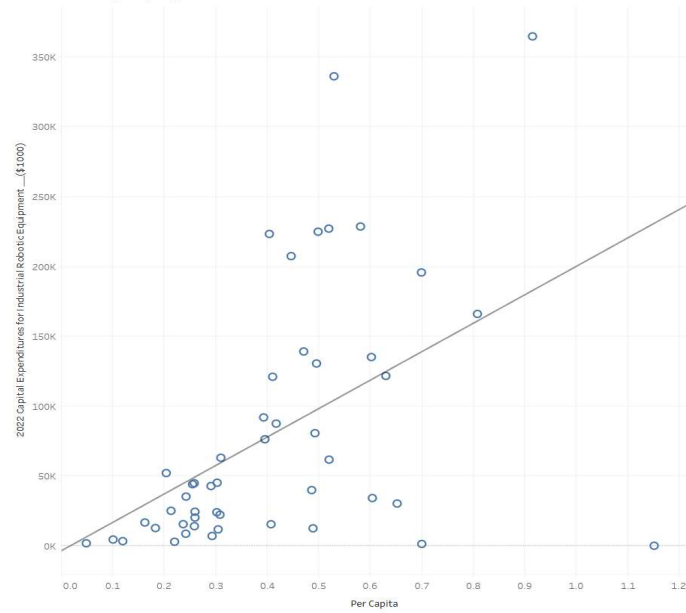
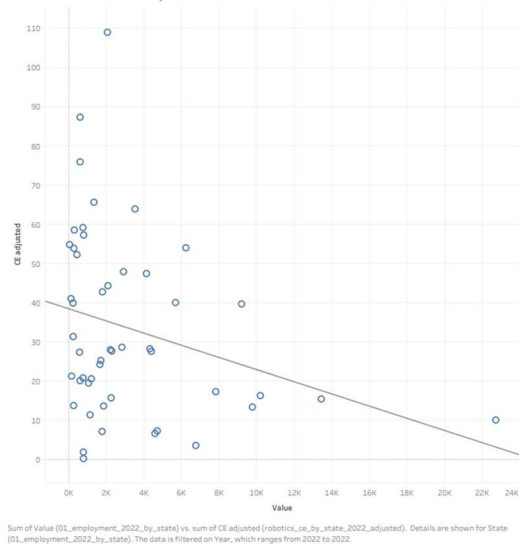


Figure 11 Exploring Investment By Total Employment

Investment Per Worker by Total Workers



2022 trends show the States that employ the most manufacturing workers tend to have had the highest Capital Expenditure in Robotics Equipment, but the least expenditure per worker.

Exploring Correlation Between Robotics Investment and Future Employment

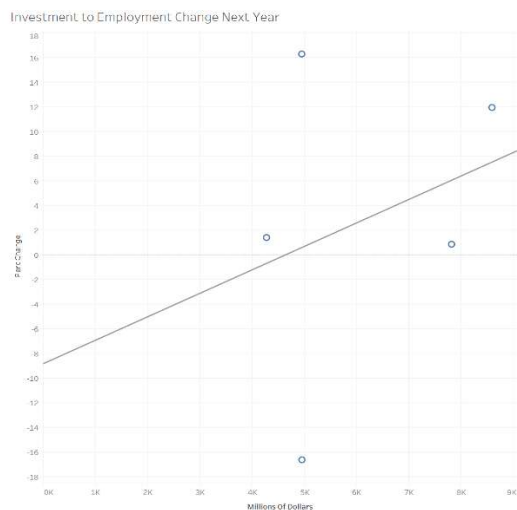


Figure 12 By Year

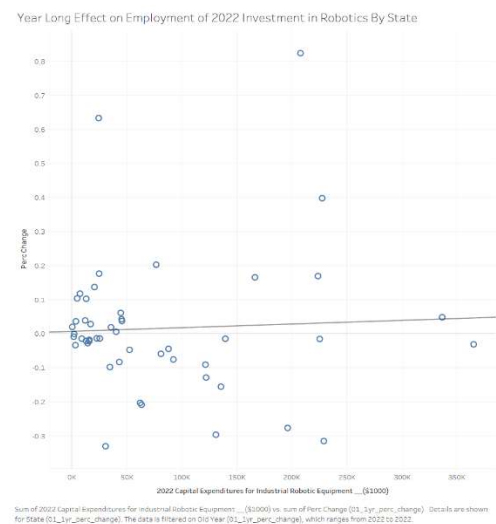


Figure 13 For 2022 By State

These trend lines show that there is a positive correlation between Robotics investment and job growth on a yearly basis. The second chart shows a weak positive correlation between investment and growth on a state-by-state level in 2022.

Conclusion

Manufacturing employment can be seen to have declined overall from the 2008 Financial Crisis and has never fully recovered. This hurt east coast workers disproportionately, especially in New York, Massachusetts, West Virginia, and North Carolina. In 2020 there is another sharp decline in total employment during Covid, along with a rise in investments seen by capital expenditure in robotics which was presumably a response by businesses to offset economic decline. Manufacturing employment returned to before Covid levels but is still less than before 2008.

There may be a weak positive correlation suggesting robotics investment supports job growth in the short term, but data is limited preventing conclusive evidence. It appears this is an area of ongoing research and policy makers find it highly relevant. This exploratory data project reaffirms existing knowledge and points to more research being needed in this area.